

Kytril® Ampoules

Granisetron

Antiemetics, Serotonin (5HT₃) antagonists

1. Pharmaceutical Form

Kytril is supplied as film-coated tablets and ampoules

2. Qualitative and Quantitative Composition

Active ingredient: granisetron HCL

Clear glass ampoules contain 1 mg of granisetron (free base equivalent) in 1 ml or 3 mg of granisetron (free base equivalent) in 3 ml. as a clear colourless liquid.

3. Clinical Particulars

3.1 Therapeutic Indications

KYTRIL is indicated for the prevention and treatment (control) of

- a) acute and delayed nausea and vomiting associated with chemotherapy and radiotherapy
- b) post-operative nausea and vomiting

3.2 Dosage and Method of Administration

3.2.1 Standard Dosage

Chemotherapy Induced Nausea and Vomiting (CINV)

Adults

Prevention: A dose of 1-3 mg (10-40mcg/kg) of Kytril should be administered either as a slow intravenous injection (over 30 seconds) or as an intravenous infusion diluted in 20 to 50 ml infusion fluid and administered over 5 minutes, prior to the start of chemotherapy.

Treatment: A dose of 1-3 mg (10-40mcg/kg) Kytril should be administered either as a slow intravenous injection (over 30 seconds) or as an intravenous infusion diluted in 20 to 50 ml infusion fluid and administered over 5 minutes. Further treatment doses of Kytril may be administered, if required, at least 10 minutes apart. The maximum dose of Kytril to be administered over 24 hours should not exceed 9 mg.

Pediatrics

Prevention and treatment: A dose of 10-40 mcg/kg body weight (up to 3 mg) should be administered as an intravenous infusion, diluted in 10 to 30 ml infusion fluid and administered over 5 minutes prior to the start of chemotherapy. One additional dose may be administered within a 24 hour period if required. This additional dose should not be administered until at least 10 minutes after the initial infusion.

Radiotherapy Induced Nausea and Vomiting (RINV)

Adults

Prevention: A dose of 1-3 mg (10-40mcg/kg) of Kytril should be administered either as a slow intravenous injection (over 30 seconds) or as an intravenous infusion diluted in 20 to 50 ml infusion fluid and administered over 5 minutes, prior to the start of radiotherapy.

Treatment: There is insufficient information to recommend the intravenous administration of KYTRIL in the treatment of RINV in adult patients.

Pediatrics

There is insufficient information to recommend intravenous administration of Kytril in the prevention and treatment of RINV in children.

Post-operative Nausea and Vomiting (PONV)

Adults

Prevention: A dose of 1 mg (10mcg/kg) of Kytril should be administered as a slow intravenous injection (over 30 seconds) prior to induction of anesthesia.

Treatment: A dose of 1 mg (10mcg/kg) of Kytril should be administered by slow intravenous injection (over 30 seconds). The maximum dose for patients undergoing anesthesia for surgery is a total dose of 3 mg of Kytril i.v. in one day.

Pediatrics

There is insufficient information to recommend intravenous administration

of Kytril in the prevention and treatment of postoperative nausea and vomiting in pediatric patients.

3.2.2 Special Dosage Instruction

Geriatrics: No dosage adjustments required.

Renal impairment: No dosage adjustments required.

Hepatic Impairment: No dosage adjustments required.

3.3 Contraindications

Kytril is contraindicated in patients with known hypersensitivity to granisetron or to any of its excipients.

3.4 Special Warnings and Special Precautions for Use

As Kytril may reduce lower bowel motility, patients with signs of subacute intestinal obstruction should be monitored closely following administration of Kytril.

As with other 5-HT₃ antagonists, cases of ECG modifications including QT prolongation have been reported with KYTRIL. These ECG changes with KYTRIL were minor and generally not of clinical significance, specifically with no evidence of proarrhythmia. However, in patients with pre-existing arrhythmias or cardiac conduction disorders, this might lead to clinical consequences. Therefore, caution should be exercised in patients with cardiac co-morbidities, on cardio-toxic chemotherapy and/or with concomitant electrolyte abnormalities.

Cross-sensitivity between 5-HT₃ antagonists has been reported.

It is recommended that Kytril film-coated tablets are not taken by patients with rare hereditary problems of galactose intolerance, lactase deficiency or glucose-galactose malabsorption.

As with other 5-HT₃ antagonists, cases of serotonin syndrome (including altered mental status, autonomic dysfunction and neuromuscular abnormalities) have been reported following the concomitant use of Kytril and other serotonergic drugs. If concomitant treatment with granisetron and other serotonergic drugs is clinically warranted, appropriate observation of this patient is advised.

3.5 Interactions with other Medicinal Products and Other Forms of Interaction

Kytril did not induce or inhibit the cytochrome P₄₅₀ drug metabolizing enzyme system in rodent studies or inhibit the activity of any well characterized P₄₅₀ sub-families studied in *in vitro* investigations.

In humans, hepatic enzyme induction with phenobarbital resulted in an increase in total plasma clearance of intravenous Kytril of approximately one-quarter. In *in vitro* human microsomal studies, ketoconazole inhibited ring oxidation of Kytril. However, given the absence of pK/pD relationship with granisetron, these changes are believed to have no clinical consequences.

Kytril has been safely administered in humans with benzodiazepines, neuroleptics and anti-ulcer medications, commonly prescribed with antiemetic treatments. Additionally, Kytril has shown no apparent drug interaction with emetogenic cancer chemotherapies.

No specific interaction studies have been conducted in anesthetized patients, but Kytril has been safely administered with commonly used anesthetic and analgesic agents. In addition, the activity of the cytochrome P₄₅₀ subfamily 3A4 (involved in the metabolism of some of the main narcotic analgesic agents) is not modified by Kytril.

As for other 5-HT₃ antagonists, cases of ECG modifications including QT prolongation have been reported with KYTRIL. These ECG changes with KYTRIL were minor and generally not of clinical significance, specifically with no evidence of proarrhythmia. However, in patients concurrently treated with drugs known to prolong QT interval and/or are arrhythmogenic, this may lead to clinical consequences.

As with other 5-HT₃ antagonists, cases of serotonin syndrome have been reported following the concomitant use of Kytril and other serotonergic drugs. If concomitant treatment with granisetron and other serotonergic drugs is clinically warranted, appropriate observation of this patient is advised (see 3.4 Special Warnings and Special Precautions for Use).

3.6 Pregnancy and Lactation

There are no studies in pregnant women and it is not known whether granisetron is excreted in human milk. Use of Kytril during pregnancy or lactation should be limited to situations where the potential benefit to the mother justifies the potential risk to the fetus or nursing infant (see 4.2.6 Preclinical Safety).

3.7 Effects on Ability to Drive and Use Machines

In healthy subjects, no clinically relevant effects on resting EEG or on the performance of psychometric tests were observed after i.v. Kytril at any dose tested (up to 200 µg/kg). There are no data on the effect of Kytril on the ability to drive or use machinery.

3.8 Undesirable Effects

3.8.1 Clinical Trials

Summary of the safety profile

The most frequently reported adverse reactions for Kytril are headache and constipation which may be transient. ECG changes including QT prolongation have been reported with Kytril (see sections 3.4 and 3.5). The following table of listed adverse reactions is derived from clinical trials and post-marketing data associated with Kytril.

Frequency categories are as follows:

Very common: $\geq 1/10$;

Common $\geq 1/100$ to $< 1/10$;

Uncommon $\geq 1/1,000$ to $< 1/100$

Rare $\geq 1/10,000$ to $< 1/1,000$

Very rare ($< 1/10,000$)

Table 1 Tabulated List of Adverse Reactions

Immune system disorders	
Uncommon	Hypersensitivity reactions e.g. anaphylaxis, urticaria
Nervous system disorders	
Very common	Headache
Uncommon	Serotonin Syndrome
Cardiac disorders	
Uncommon	QT prolongation
Gastrointestinal disorders	
Very common	Constipation
Hepatobiliary disorders	
Common	Elevated hepatic transaminases*
Skin and subcutaneous tissue disorders	
Uncommon	Rash

*Occurred at a similar frequency in patients receiving comparator therapy

Kytril has been well tolerated in human studies. In common with other drugs of this class, headache and constipation have been reported. Cases of hypersensitivity reactions, including rashes and anaphylaxis have been reported. Elevations in hepatic transaminases have been observed and at similar frequency in patients receiving comparator therapy.

As for other 5-HT₃ antagonists, cases of ECG modifications including QT prolongation have been reported with KYTRIL. These ECG changes with KYTRIL were minor and generally not of clinical significance, specifically with no evidence of proarrhythmia.

As with other 5-HT₃ antagonists, cases of serotonin syndrome (including altered mental status, autonomic dysfunction and neuromuscular abnormalities) have been reported following the concomitant use of Kytril and other serotonergic drugs (see 3.4 Special Warnings and Special Precautions for Use and 3.5 Interactions with other Medicinal Products and other Forms of Interaction).

3.8.2 Post Marketing

The post-marketing safety experience in over 4 million patients is consistent with the clinical trial safety information.

For ECG modifications, see 3.8 Undesirable Effects, Clinical Trials.

For serotonin syndrome, see 3.8 Undesirable Effects, Clinical Trials.

3.9 Overdose

There is no specific antidote for Kytril. In the case of overdosage with Kytril, symptomatic treatment should be given. Overdosage of up to 38.5 mg of granisetron hydrochloride as a single injection has been reported without symptoms or only the occurrence of a slight headache.

4. Pharmacological Properties & Effects

4.1 Pharmacodynamic Properties

4.1.1 Mechanism of Action

Serotonin receptors of the 5-HT₃ type are located peripherally in vagal nerve terminals and centrally in the chemoreceptor trigger zone of the area postrema. During chemotherapy-induced vomiting, mucosal enterochromaffin cells release serotonin, which stimulates 5-HT₃ receptors. This invokes vagal afferent discharge, inducing vomiting. Kytril is a potent anti-emetic and highly selective antagonist of 5-hydroxytryptamine (5-HT₃) receptors. Radioligand binding studies have demonstrated that Kytril has negligible affinity for other receptor types including 5-HT and dopamine D₂ binding sites.

4.1.2 Efficacy / Clinical Studies

Chemotherapy-induced nausea and vomiting (CINV)

KYTRIL administered intravenously has been shown to be effective in the prevention and treatment of nausea and vomiting associated with cancer chemotherapy in adults. Kytril administered orally has been shown to be effective in the prevention of nausea and vomiting associated with cancer chemotherapy. Kytril administered intravenously has been shown to be effective in children aged 2 years and above for the prevention and treatment of acute CINV. There is insufficient information to recommend the oral administration of KYTRIL in the prevention and treatment of CINV in pediatric patients.

Radiation-induced nausea and vomiting (RINV)

Kytril administered orally has been shown to be effective in the prevention and treatment of nausea and vomiting associated with total body or fractionated abdominal irradiation in adults. Efficacy in children has not been established in controlled clinical trials.

Postoperative nausea and vomiting (PONV)

Kytril administered intravenously has been shown to be effective for prevention and treatment of post-operative nausea and vomiting in adults.

4.2 Pharmacokinetic Properties

4.2.1 Absorption

Absorption of Kytril is rapid and complete, though oral bioavailability is reduced to about 60% as a result of first pass metabolism. Oral bioavailability is generally not influenced by food.

4.2.2 Distribution

Kytril is extensively distributed, with a mean volume of distribution of approximately 3l/kg. Plasma protein binding is approximately 65%.

4.2.3 Metabolism

Biotransformation pathways involve N-demethylation and aromatic ring oxidation followed by conjugation. In-vitro liver microsomal studies show that granisetron's major route of metabolism is inhibited by ketoconazole, suggestive of metabolism mediated by the cytochrome P-450 3A subfamily.

4.2.4 Elimination

Clearance is predominantly by hepatic metabolism. Urinary excretion of unchanged Kytril averages 12% of dose while that of metabolites amounts to about 47% of dose. The remainder is excreted in feces as metabolites. Mean plasma half-life in patients by the oral and intravenous route is approximately 9 hours, with a wide inter-subject variability. The pharmacokinetics of oral and intravenous Kytril demonstrate no marked deviations from linear pharmacokinetics at oral doses up to 2.5-fold and intravenous doses up to 4-fold the recommended clinical dose. The results of a study in healthy male volunteers have demonstrated that systemic delivery of 3 mg granisetron from an intramuscular injection is slower than from a 5 minute intravenous infusion (as indicated by lower C_{max} and later T_{max}). In other respects, the pharmacokinetics of granisetron are virtually indistinguishable when administered by these two different routes.

4.2.5 Pharmacokinetics in Special Populations

Renal failure: In patients with severe renal failure, data indicate that pharmacokinetic parameters after a single intravenous dose are generally similar to those in normal subjects.

Hepatic impairment: In patients with hepatic impairment due to neoplastic liver involvement, total plasma clearance of an intravenous dose was approximately halved compared to patients without hepatic involvement. Despite these changes, no dosage adjustment is necessary.

Elderly: In elderly subjects after single intravenous doses, pharmacokinetic parameters were within the range found for non-elderly subjects.

Pediatrics: In children, after single intravenous doses, pharmacokinetics are similar to those in adults when appropriate parameters (volume of distribution, total plasma clearance) are normalized for body weight.

4.2.6 Preclinical Safety

Preclinical data revealed no special hazard for humans based on conventional studies of safety pharmacology, repeated dose toxicity, reproductive toxicity and genotoxicity. Carcinogenicity studies revealed no special hazard for humans when used in the recommended human dose.

However, when administered in higher doses and over a prolonged period of time the risk of carcinogenicity cannot be ruled out.

In carcinogenicity studies in rats and mice treated orally for their lifetime (2 years), no adverse findings were observed at dosages 25 times the clinical dose. At higher doses, KYTRIL induced cell proliferation in the rat liver and hepatocellular tumors in rats and mice.

KYTRIL was not mutagenic in mammalian or non-mammalian in vivo or in vitro test systems and there was no evidence of unscheduled DNA synthesis indicating that KYTRIL is not genotoxic.

In the rat, KYTRIL had no untoward effect on reproductive performance, fertility or on pre- and post natal development. Teratogenic effects were not observed in rats or rabbits.

5. Pharmaceutical Particulars

5.1 List of Excipients

Ampoules:

Sodium chloride

Water for injections Ph. Eur.

5.2 Incompatibilities

Admixtures of granisetron hydrochloride and dexamethasone sodium phosphate are compatible at concentrations of 10 to 60 mcg/ml granisetron and 80 to 480 mcg/ml dexamethasone phosphate in either 0.9% sodium chloride or 5% glucose intravenous infusion fluids.

5.3 Stability

This medicinal product should not be used after the expiry date (EXP) shown on the outer pack.

5.4 Special Remarks

Admixtures of granisetron hydrochloride and dexamethasone sodium phosphate are compatible at concentrations of 10 to 60 µg/ml granisetron and 80 to 480 µg/ml dexamethasone phosphate in either 0.9% sodium chloride or 5% glucose intravenous infusion fluids.

The admixture will have a shelf-life of 24 hours.

5.4.1 Special Precautions for Storage

Ampoules, protect from light

Do not store above 30°C, keep the ampoule in outer carton.

See also outer pack for storage remarks.

5.4.2 Instructions for Use, Handling and Disposal

Preparation of Infusion

For adults: The appropriate dose is diluted with infusion fluid, to a total volume of 20 to 50ml in any of the following solutions: 0.9% sodium chloride, 0.18% sodium chloride and 4% dextrose, 5% dextrose, Hartmann's solution, sodium lactate and mannitol.

For children: The appropriate dose is diluted with infusion fluid (as for adults) to a total volume of 10 to 30ml.

Kytril has been shown to be stable for at least 24 hours when stored at ambient temperature in any of the following solutions: 0.9% sodium chloride, 0.18% sodium chloride and 4% dextrose, 5% dextrose, Hartmann's solution, sodium lactate and mannitol.

From a microbiological point of view the diluted product should be used immediately. If not used immediately, in use storage times and conditions prior to use are the responsibility of the user and would normally not be longer than 24 h at 2°C – 8°C, unless dilution has taken place in controlled and validated aseptic conditions.

6. Packs

Ampoules, 1mg/1ml, 3mg/3ml	5, 5
Tablets, 1mg	2, 10

Medicine: Keep out of reach of children

MYKytrilAmp0714/CDS3.0



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Ampoules

Made for

Atmahs Pharma UK Ltd,

Basildon, United Kingdom.

By CENEXI SAS,

Fontenay-sous-Bois, France.